

Ex 13a / Page 775

$$f(x) = \frac{1}{(1+x)^2}$$

$$\int f(x) = -\frac{1}{1+x} + C = -\frac{1}{1-(-x)} + C$$

$$-\frac{1}{1-(-x)} = -(1-x+x^2-x^3+\dots) = -1+x-x^2+x^3\dots$$

$$\left(-\frac{1}{1+x} + C\right)' = \frac{1}{(1+x)^2}$$

$$\left(-1+x-x^2+x^3-x^4+x^5+\dots\right)' = 1-2x+3x^2-4x^3+\dots$$

$$\frac{1}{1+x^2} = 1-2x+3x^2-4x^3+\dots = \sum_{n=0}^{\infty} (-1)^n (n+1) x^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left(\frac{(-1)^{n+1} \cdot (n+2) (x^{n+1})}{(-1)^n (n+1) \cdot x^n} \right) =$$

$$= \lim_{n \rightarrow \infty} \left| -\frac{(n+2)x}{n+1} \right| = \lim_{n \rightarrow \infty} \frac{(n+2)|x|}{n+1} = |x| > 1$$

$$R=1 \quad -1 < x < 1$$

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$$f(x) = x^4 - 3x^2 + 1$$

$$a=1$$

$$f(a) = 1 - 3 + 1 = -1$$

$$f'(a) = -2$$

$$f''(a) = 6$$

$$f'''(a) = 24$$

$$f^{(4)}(a) = 24$$

$$f^{(5)}(a) = 0$$

$$f(x) = \frac{f(a)}{0!} + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots$$

$$f(x) = -1 + \frac{-2}{1} (x-1) + \frac{6}{2} (x-1)^2 + \frac{24}{6} (x-1)^3 + \frac{24}{24} (x-1)^4 =$$

$$= -1 - 2(x-1) + 3(x-1)^2 + 4(x-1)^3 + (x-1)^4$$

$$\forall x \text{ we get } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 ; R = \infty$$

Ex 231 / Page 131

$$f(x) = e^x + e^{2x}$$

$$f(0) = 2$$

$$f'(0) = 3$$

$$f''(0) = 5$$

$$f'''(0) = 7$$

$$f(x) = 2 + \frac{3x}{1} + \frac{5x^2}{2} + \frac{7x^3}{6} + \frac{8x^4}{24} + \dots = \sum_{n=0}^{\infty} \frac{2^{n+1}}{n!} x^n$$

Ex 237 / Page 780

$$\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{1}{6}x^3}{x^5}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\lim_{x \rightarrow 0} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) - x + \frac{1}{6}x^3$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots}{x^5} = \lim_{x \rightarrow 0} \left(\frac{1}{5!} - \frac{x^2}{7!} + \frac{x^4}{9!} - \dots \right) =$$

$$= \frac{1}{5!} = \frac{1}{120}$$

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$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n}}{n!} = 1 - \frac{x^4}{1} + \frac{x^8}{2} - \frac{x^{12}}{6} + \frac{x^{16}}{24} =$$

$$e^{-x^4} = 1 + \frac{-x^4}{1} + \frac{(-x^4)^2}{2!} - \frac{(-x^4)^3}{3!} + \dots$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{x^{4n}}{n!} = e^{-x^4}$$